

spot checkers lis alt proof

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0 big can be checked in $\log n$ time. candidate 2^n being witness for i being $1/4$ big is checked in $O(1)$ (this overflows but that's ok) assume that i is 0-big. check s elements in interval $[i, i + 2^n]$. the probability that a witness j in the interval $[i + 2^{n-1}, i + 2^n]$ is NOT found is $(3/4)^s$.

check $1/\varepsilon$ times. check for 0 big. if not found, continue. else fail.

pass

assume: ε violate monotonicity

Lemma 0.1. *if an element i is in a subset s such that when $n - s$ gives an optimal monotone subsequence, then i is 0 big.*

Proof. i must have at least one witness j such that $i < j, f(i) > f(j)$ or $i > j, f(i) < f(j)$. □

Lemma 0.2. *given an element in a list length n and that is $\varepsilon_f n$ away from monotonicity there exists an algorithm detects its violation from monotonicity that runs in $\log n$ time*

Proof. □